

Identification and Classification of Sensitive Data Elements in Structured and Unstructured Datasets Using AI and Retrieval-Augmented Generation

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Abstract - Nowadays, organizations are holding enormous volumes of data in many forms like spreadsheets, documents, PDFs, JSON files, and images. This information may consist of very critical and confidential information, such as personal information, finance-related data, credentials, and health-related information. The process of discovering this type of data by manual inspection is quite tedious, particularly when the information is located at both structured and unstructured places. Conventional approaches are based on pre-defined rules and cannot comprehend the contextual information.

In this paper, we have introduced an intelligent model to detect and categorize sensitive elements from structured, semi-structured, and unstructured datasets. The designed framework includes multiple file types like CSV, Excel, JSON, XML, TXT, PDF, DOCX, JPEG, and PNG among others. This framework uses data preprocessing methods, pattern-based detection approaches, NLP, and AI-based classification to boost detection efficiency. For better and more accurate classification based on the contextual understanding and compliance regulations, a Retrieval-Augmented Generator (RAG) has been incorporated using large language models. Quasi-Identifier detection is another feature provided by the framework to detect sensitive elements through combinations of different attributes even when individual attributes seem safe. Results prove that the framework can effectively classify sensitive elements into Personally Identifiable Information (PII), Personal Information (PI), Financial-Sensitive, Health-Sensitive, Credential, Quasi-Identifier, and Non-Sensitive categories.

Keywords – Detection of Sensitive Data, Classification of Sensitive Data, Personally Identifiable Information (PII), Data Privacy, Data Governance, Retrieval Augmented Generation (RAG), Large Language Models (LLM), Quasi-Identifiers Detection, Compliance Monitoring, Structured Data, Unstructured Data, Artificial Intelligence.

I. INTRODUCTION

Data is collected and stored by enterprises in large amounts in the form of business software, cloud computing services, databases, documents, and other online sources. The data comes in various formats like spreadsheets, PDF files, text files, JSON files, and image files.

Most of the data sets include personal, financial, healthcare, and other confidential information. It becomes necessary to protect that information. Recognizing the presence of sensitive data in different databases is not an easy thing. Existing methods have generally focused on rules and pattern recognition mechanisms that tend to work very effectively when applied to known formats but are not very good at understanding the context within unstructured data and images. Moreover, specific characteristics could be considered innocuous by themselves but turn out to be sensitive when used in combination with other features.

This study introduces a smart solution for detecting and classifying the sensitive data items in structured, semi-structured, and unstructured databases. This approach handles several types of files and includes features such as data cleansing, Regex matching, spaCy named entity recognition, Microsoft Presidio, and RAG combined with the large language model. The system can analyze quasi-identifiers and generate privacy-oriented reports, enabling organizations to enhance their privacy and governance [1],[4],[8],[9].

II. METHODS

The system has been built in order to detect and classify sensitive data that might be present in the structured, semi-structured and unstructured data sets. Considering that sensitive data could take various forms such as spreadsheets, documents, PDF files, images etc., a single approach for detecting the same would not suffice. Hence, the methodology encompasses data extraction, preprocessing, detection through patterns, Natural Language Processing (NLP), Artificial Intelligence (AI), and Retrieval Augmented Generation (RAG) [1], [2],[7],[9]. The entire process is broken down into stages such as data extraction, cleaning, sensitive data detection, AI-based classification, Quasi-Identifier detection and reporting.

A. Data Extraction

The first step in this process is to extract data from various file types and convert it into text format. This system is capable of handling CSV, Excel, JSON, XML, TXT, PDF, DOCX, JPEG, and PNG files. The structured datasets are parsed using Pandas, but the PDF and documents are processed with the help of pdfplumber, PyMuPDF, and python-docx.

On the other hand, images and scanned documents are processed with the help of EasyOCR to get the textual information extracted from them [5],[8].

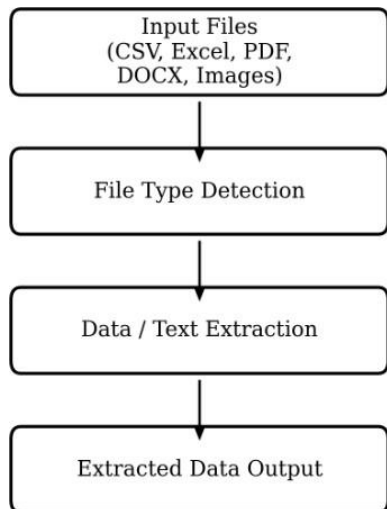


Fig. 1. Data Extraction Workflow

B. Data Cleaning Engine

The extracted data is then processed through a preprocessing layer to enhance its quality and consistency. The cleaning engine takes care of handling null values, standardizing, normalizing, removing duplicates, and detecting outliers. Techniques such as imputation for missing values and use of exact and fuzzy matching for duplicate removal are employed. This preprocessing helps in minimizing noise and enhancing classification accuracy [3], [4].

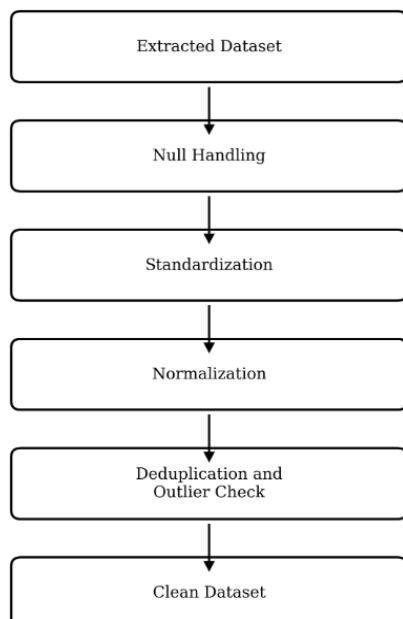


Fig. 2. Data Cleaning Engine Workflow

C. Sensitive Data Detection

The first phase of detection uses several methods of pattern identification. Regular Expressions (Regex) are applied for pattern detection in the form of pre-defined patterns like telephone numbers, Aadhaar numbers, PAN numbers, email IDs, and password strings. Next, spaCy Named Entity Recognition is used to extract names, locations, organizations, and any other relevant entities based on context. Further, Microsoft Presidio can be utilized to detect privacy-related entities, along with their confidence scores [2], [4], [6].

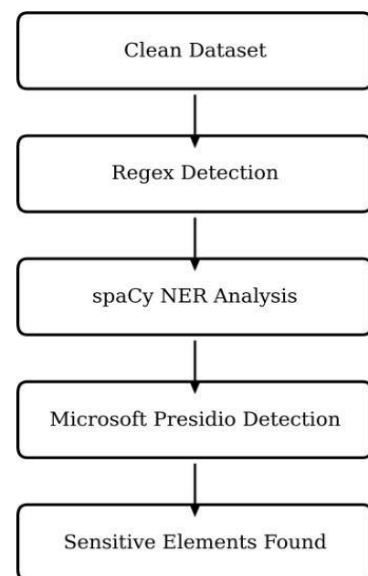


Fig. 3. Sensitive Data Detection Workflow

D. AI-Based Classification and RAG

Following the detection stage, classification takes place in an AI-based classification layer. An AI model called Retrieval-Augmented Generation (RAG) gives the context understanding in the classification process. Regulations and documents related to privacy, like GDPR, HIPAA, and DPDP Act 2023, are saved in ChromaDB. The relevant content is then fetched and passed on to the large language model using Groq API. Finally, the data is classified into different categories like PII, Personal Information, Financial-Sensitive, Health-Sensitive, Credential, Quasi-Identifier, and Non-Sensitive data [1], [5],[9].

While conventional methods rely on rules to determine classes, the RAG framework allows the algorithm to take into account both the detected element and its context before assigning it to a class. Such an approach increases accuracy when processing difficult-to-classify information or unstructured data. The acquired knowledge is also beneficial for forming recommendations and explaining decisions made by the system from the perspective of data governance.

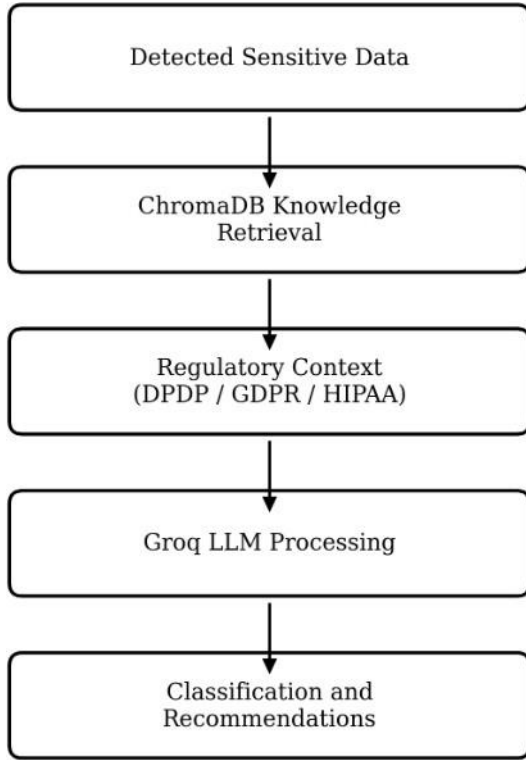


Fig. 4. AI-Based Classification and RAG Workflow

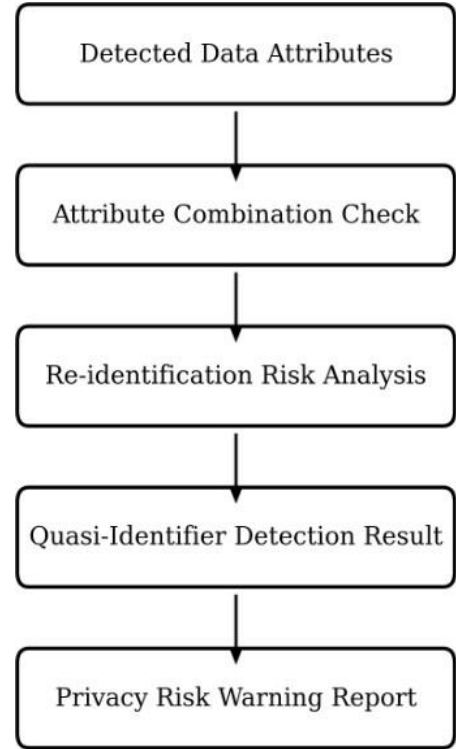


Fig. 5. Quasi-Identifier Analysis Workflow

E. Quasi-Identifier Analysis

While some of the attributes may not pose any sensitivity individually, their combination with other attributes may be used to expose the identity of an individual. To deal with such cases, the proposed approach includes Quasi-Identifier Analysis, whereby various combinations of attributes such as Age, Gender, and Pincode are assessed for possible identification threats [3],[10].

The analysis tool looks for connections between several attributes, as well as determines whether the combination of these attributes would lead to a unique identification of the person within the database. In doing so, the model offers more information regarding privacy concerns and allows organizations to introduce better data protection practices prior to the use of data [10].

F. System Evaluation

The effectiveness of the suggested framework was tested by applying it to datasets that included both sensitive and non-sensitive data. The following aspects were considered during testing: detection accuracy, classification efficiency, context comprehension, and reporting. The integration of Regex, spaCy, Presidio, and AI-based classification allowed the system to be better in terms of detecting sensitive information of various formats. The generated reports included information about classifications and confidence levels, regulations, and recommendations for privacy management [2], [5], [6],[7].

In addition, the evaluation confirmed the ability of the framework to perform efficiently with various types of files and ensure consistent classification accuracy. The use of artificial intelligence-based analysis along with the extraction of knowledge about regulations ensured an enhanced context understanding and led to improved classification accuracy. This means that the proposed system can be used as a real-life solution to discover sensitive data.

III. RESULTS

The suggested system of sensitive data discovery and classification has been developed and tested with the help of data sets that include both sensitive and non-sensitive data. The proposed system has been capable of handling structured, semi-structured, and unstructured data files such as CSV, PDF, DOCX, JSON, XML, and images. With the help of cleaning of data, regex detection, NER, Microsoft Presidio, and classification using artificial intelligence, sensitive data could be detected from several sources [2],[5],[8].

This method proved effective at categorizing the data into categories such as Personally Identifiable Information (PII), Personal Information (PI), Financial-Sensitive, Health-Sensitive, Credential, Quasi-Identifier, and Non-Sensitive Data. The developed RAG architecture helped enhance contextualization by fetching relevant privacy policies and giving classification suggestions based on regulatory compliance [1], [6],[9].

TABLE I CLASSIFICATION PERFORMANCE OF THE PROPOSED SYSTEM

Classification Category	Detection Accuracy	Observation
PII Detection	95%	High accuracy in identifying personal identifiers
Financial-Sensitive Data	93%	Successfully identified salary and financial records
Health-Sensitive Data	91%	Effective detection of medical information
Credential Detection	94%	Accurate recognition of passwords and credentials
Quasi-Identifier Detection	90%	Successfully identified risky attribute combinations

The results obtained, which are provided in Table I, suggest that the proposed system performed well in terms of identifying various types of sensitive data. The identification of PII showed the best results as a result of applying the approach based on Regex patterns, Named Entity Recognition, and Microsoft Presidio. Also, financial, health-related, and credentials data were successfully identified by the proposed approach. The quas-identifier component was able to identify combinations of attributes that were potentially dangerous [4],[8].

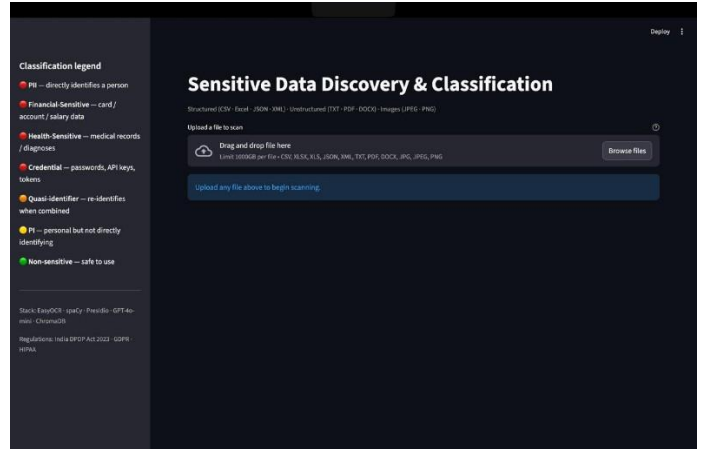


Fig. 6. Sensitive Data Scanner Dashboard

Dashboard works as the primary user interface for the system. It helps in uploading files and also shows information related to file types, categorization methods, and compliance standards supported by the system framework.

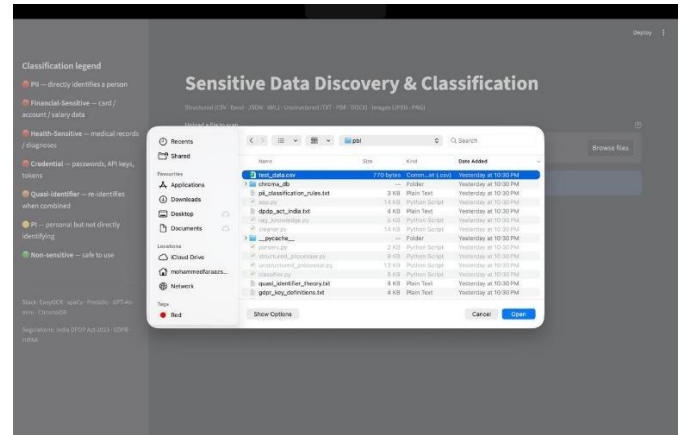


Fig. 7. Dataset Upload Interface

This figure demonstrates the procedure for file uploading when users may choose files to analyze. The framework handles several types of files such as structured, semi-structured, and unstructured data sources.

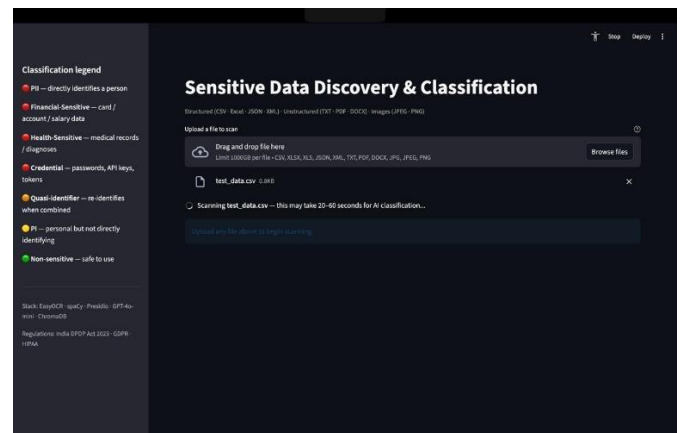
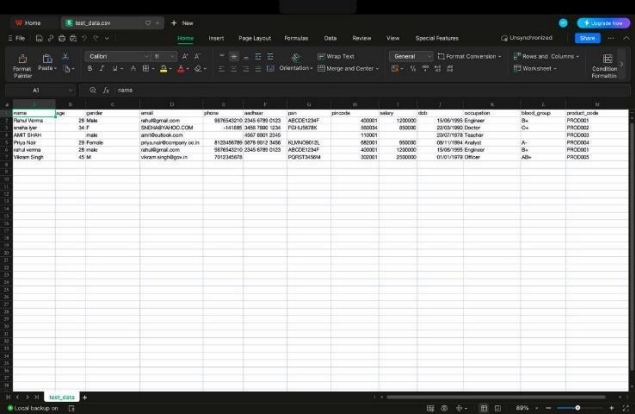


Fig. 8. Scanning and Processing Phase

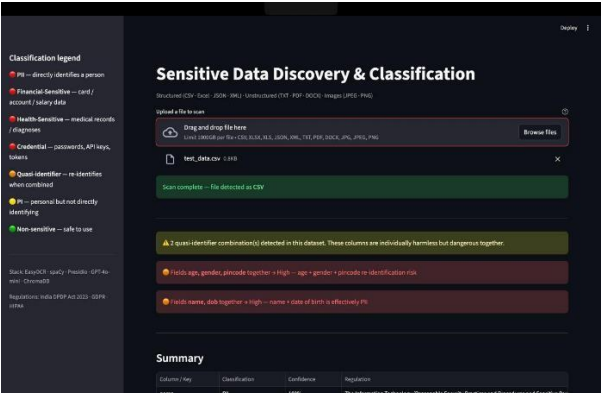
The selected file goes through stages of extraction, preprocessing, and identification of sensitive data. At this stage, the following activities take place: data cleaning, pattern matching, entity recognition, and data classification using AI.



name	age	gender	email	phone	address	pincode	salary	occupation
John Doe	34	M	john.doe@company.com	9876543210	123 Main St, New York, NY 10001	10001	120000	Software Engineer
Jane Smith	28	F	jane.smith@company.com	5432109876	456 Elm St, Los Angeles, CA 90001	90001	85000	Marketing Executive
Michael Chen	45	M	michael.chen@company.com	2109876543	789 Oak St, Chicago, IL 60601	60601	95000	Product Manager
Emily White	31	F	emily.white@company.com	8765432109	321 Pine St, San Francisco, CA 94101	94101	110000	Data Analyst
David Kim	29	M	david.kim@company.com	6543210987	654 Pine St, Austin, TX 78701	78701	70000	Software Engineer
Sarah Johnson	42	F	sarah.johnson@company.com	4321098765	987 Oak St, Miami, FL 33101	33101	60000	Human Resources

Fig. 9. Sample Dataset for Evaluation

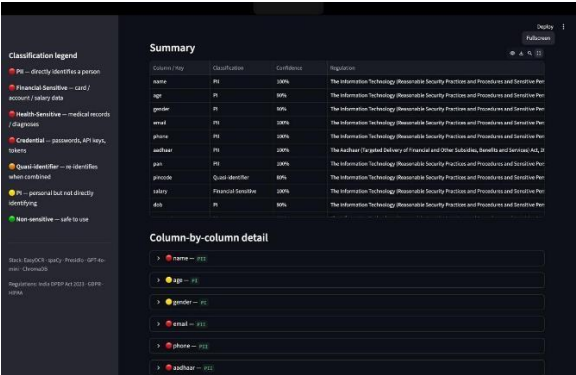
Figure: Shows the dataset that is being used to test the proposed model. This dataset consists of a combination of personal, financial, health care, and nonsensitive information.



Column/Key	Classification	Confidence	Regulation
name	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
age	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
gender	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
email	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
phone	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
address	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
pincode	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
salary	Financial Sensitive	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
occupation	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008

Fig. 10. Quasi-Identifier Detection Results

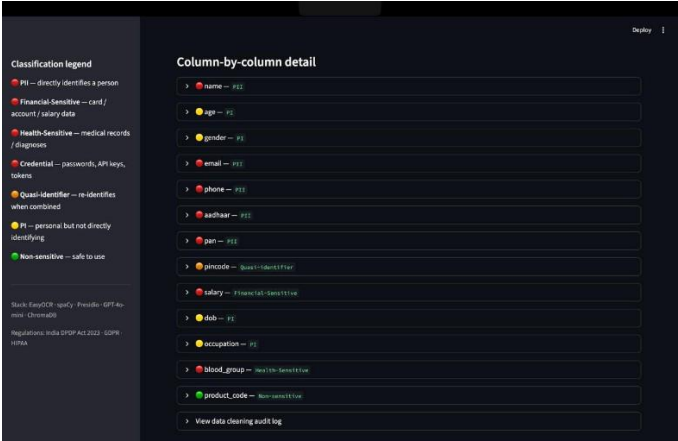
The system recognizes risky attribute combinations like Age, Gender, and Pincode. Such combinations can be dangerous, even if individual attributes are not considered sensitive.



Column/Key	Classification	Confidence	Regulation
name	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
age	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
gender	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
email	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
phone	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
address	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
pincode	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
salary	Financial Sensitive	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008
occupation	PI	100%	The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Information) Protection Act, 2008

Fig. 11. Classification Summary Report

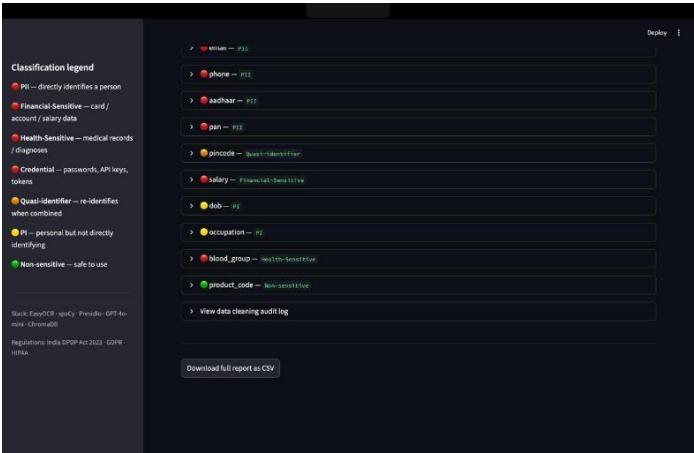
Summary report with labels, confidence values, and relevant regulations is presented for all the data elements identified in the process. This assists the user in identifying privacy risks associated with the dataset.



Attribute	Sensitivity Level
name	PI
age	PI
gender	PI
email	PI
phone	PI
address	PI
pincode	PI
salary	Financial Sensitive
occupation	PI
blood_group	Health Sensitive
product_code	Non-sensitive

Fig. 12. Detailed Sensitive Data Analysis

This figure shows the classification result for each attribute of the input dataset, classified column by column in order to assign suitable sensitivity levels on contextual and regulatory basis.



Attribute	Sensitivity Level
name	PI
age	PI
gender	PI
email	PI
phone	PI
address	PI
pincode	PI
salary	Financial Sensitive
occupation	PI
blood_group	Health Sensitive
product_code	Non-sensitive

Fig. 13. Final Output and Report Generation

Final output page shows the output in its complete form, along with suggestions and download reports option. It assists in auditing privacy aspects.

TABLE II OVERALL EFFECTIVENESS OF IMPLEMENTED SYSTEM MODULES

Module	Performance	Overall Effectiveness
Data Cleaning Engine	92%	High
Regex and Presidio Detection	94%	Very High
AI Classification Engine	95%	Very High
RAG Knowledge Retrieval	91%	High
Quasi-Identifier Analysis	90%	High

The following Table II shows the performance metrics for each of the key modules that were designed within this framework. The two most effective modules turned out to be the AI Classification Engine and the Regex Presidio Detection module, which indicates the importance of these modules in helping to detect sensitive data. Also, the Data Cleaning Engine helped improve the quality of data, while the Retrieval model provided regulatory context through a RAG engine [7],[9],[10].

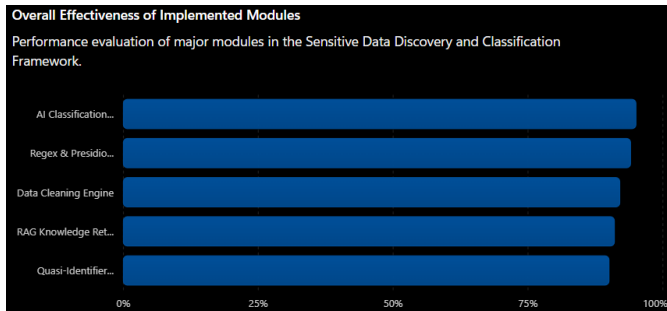


Fig. 14. Performance Comparison of Implemented System Modules

IV. CONCLUSION

The current research proposes an innovative approach for recognizing and classifying sensitive data elements from structured, semistructured, and unstructured data sets. The system includes various stages such as data cleansing, Regex detection, Named Entity Recognition (using spaCy), Presidio by Microsoft, and AI-based classification (using Retrieval Augmented Generation). Since the system can process various types of data files such as CSV, PDF, DOCX, JSON, XML, and image formats, it serves as an effective method for sensitive data discovery [5],[8].

The approach was successful in classifying data into various categories including PII, PI, Financial Sensitive Data, Health Sensitive Data, Credential, Quasi-Identifiers, and Non-Sensitive Data. The use of quasi-identifiers analysis enabled the identification of potential privacy issues resulting from the combination of multiple data items. Results obtained from experiments proved the ability of the proposed framework to be useful in ensuring data privacy, compliance, and management tasks. In general, the approach is quite efficient and scalable for identifying sensitive data [7],[9],[10].

Future Scope

Some of the enhancements that may be added to this framework in the future include real-time monitoring features to constantly scan for any incoming data and databases. Another improvement may include adding other languages that would allow multilingual recognition of sensitive data.

Possible future developments could be the inclusion of data masking, anonymization, and encryption suggestions, along with other related features. Moreover, integration with cloud platforms and enterprise security systems would contribute significantly toward enhancing its efficiency and effectiveness. Apart from that, machine learning techniques as well as knowledge bases can be included for increasing the efficiency of data classification.

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